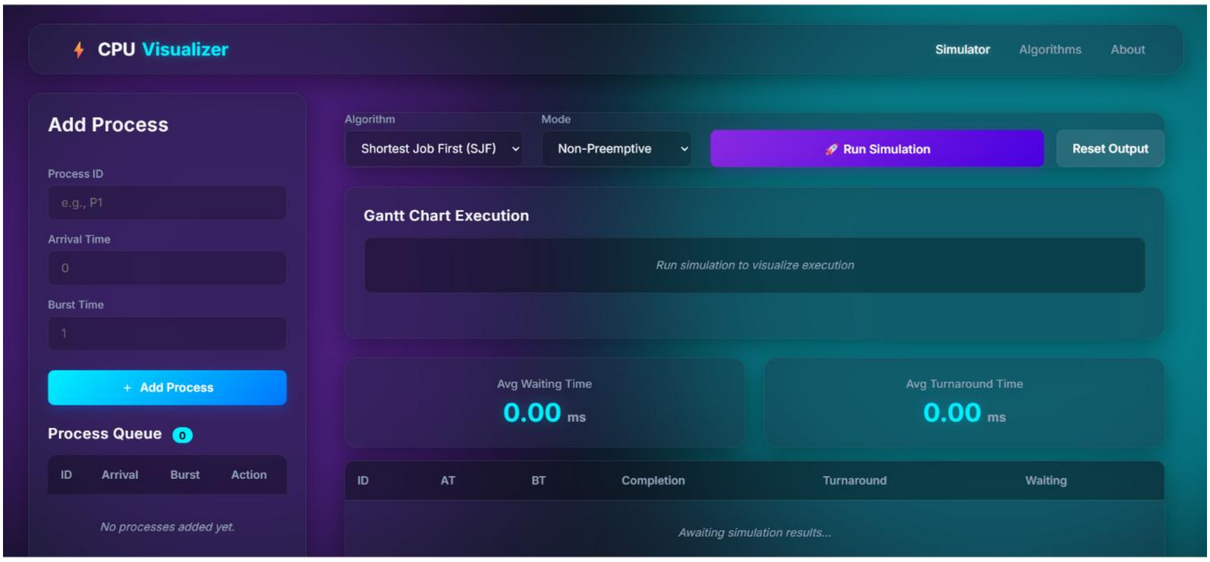
OPEN ENDED

AIM:

To design and develop a CPU Scheduling Visualizer that simulates various scheduling algorithms and visually represents process execution along with performance metrics such as waiting time and turnaround time.

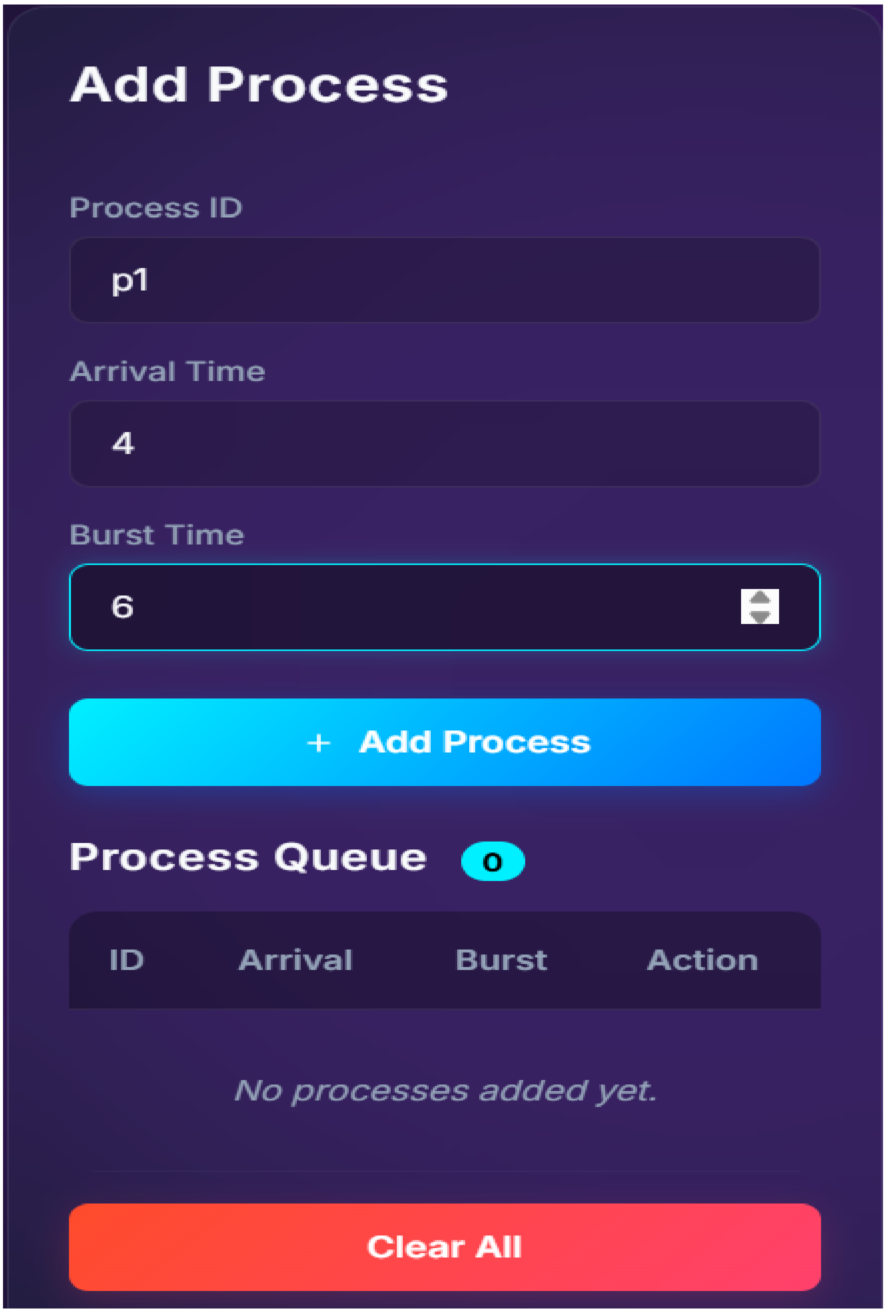
Step 1: Application Interface Initialization

This step shows the initial interface of the CPU Visualizer application. The system is in its default state where no processes have been added. The user is provided with options to enter process details, select scheduling algorithms, and run the simulation.



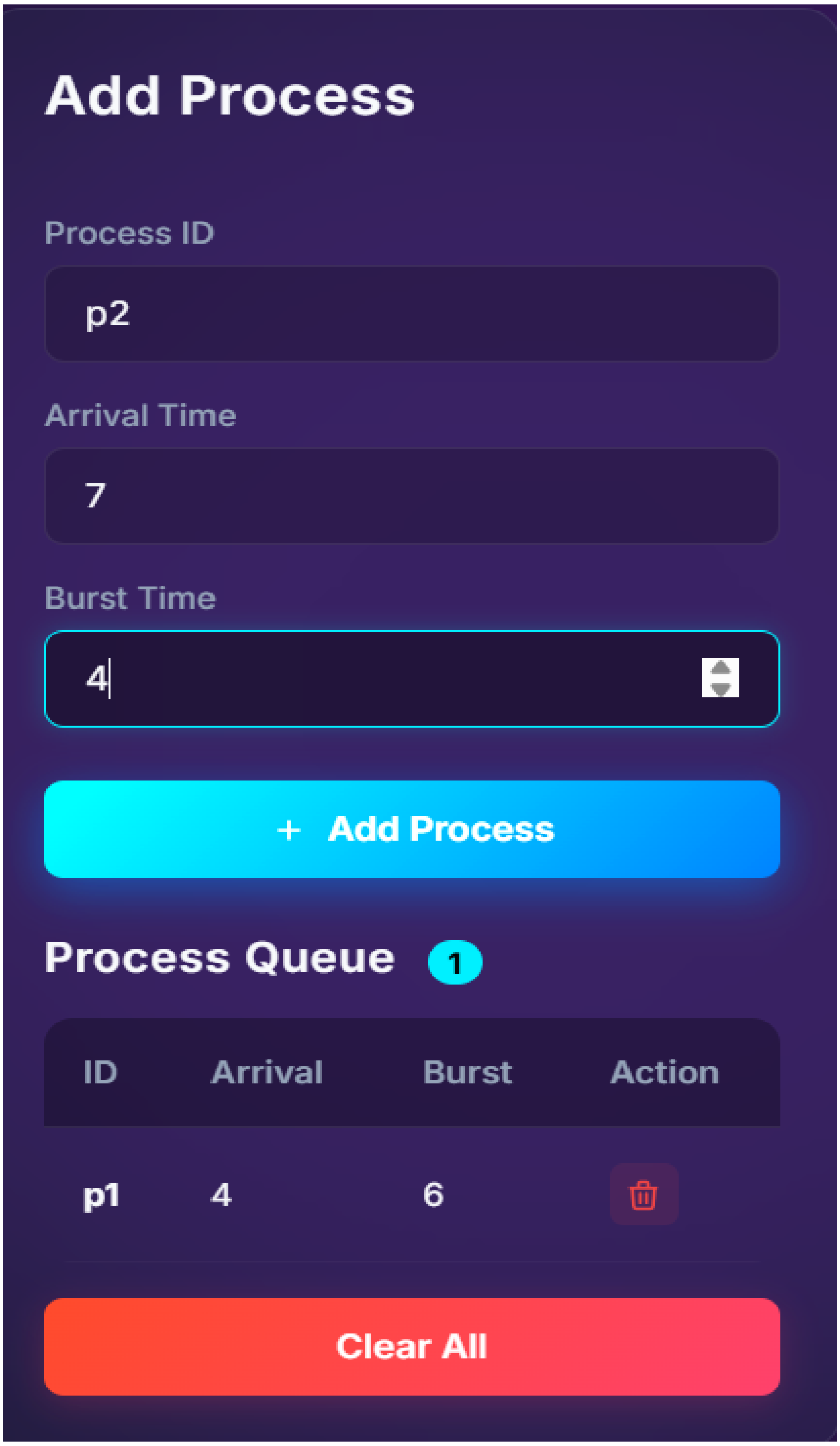
Step 2: Adding First Process (P1)

In this step, the first process **P1** is added by entering its Process ID, Arrival Time (4), and Burst Time (6). After clicking the "Add Process" button, the process is prepared to be added to the queue.



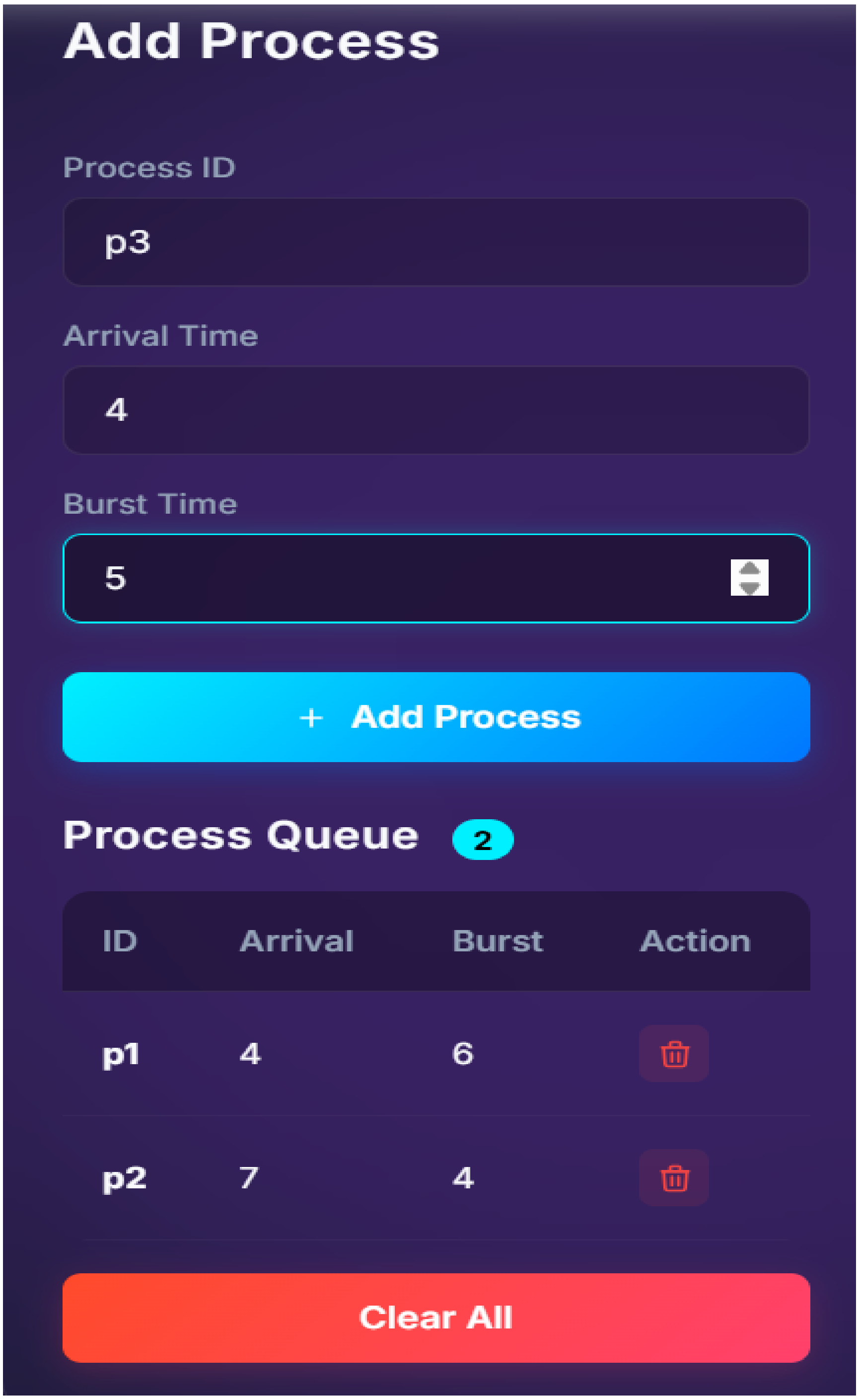
Step 3: Adding Second Process (P2)

The second process **P2** is entered with Arrival Time (7) and Burst Time (4). The system starts building a queue of processes, allowing multiple entries for simulation.



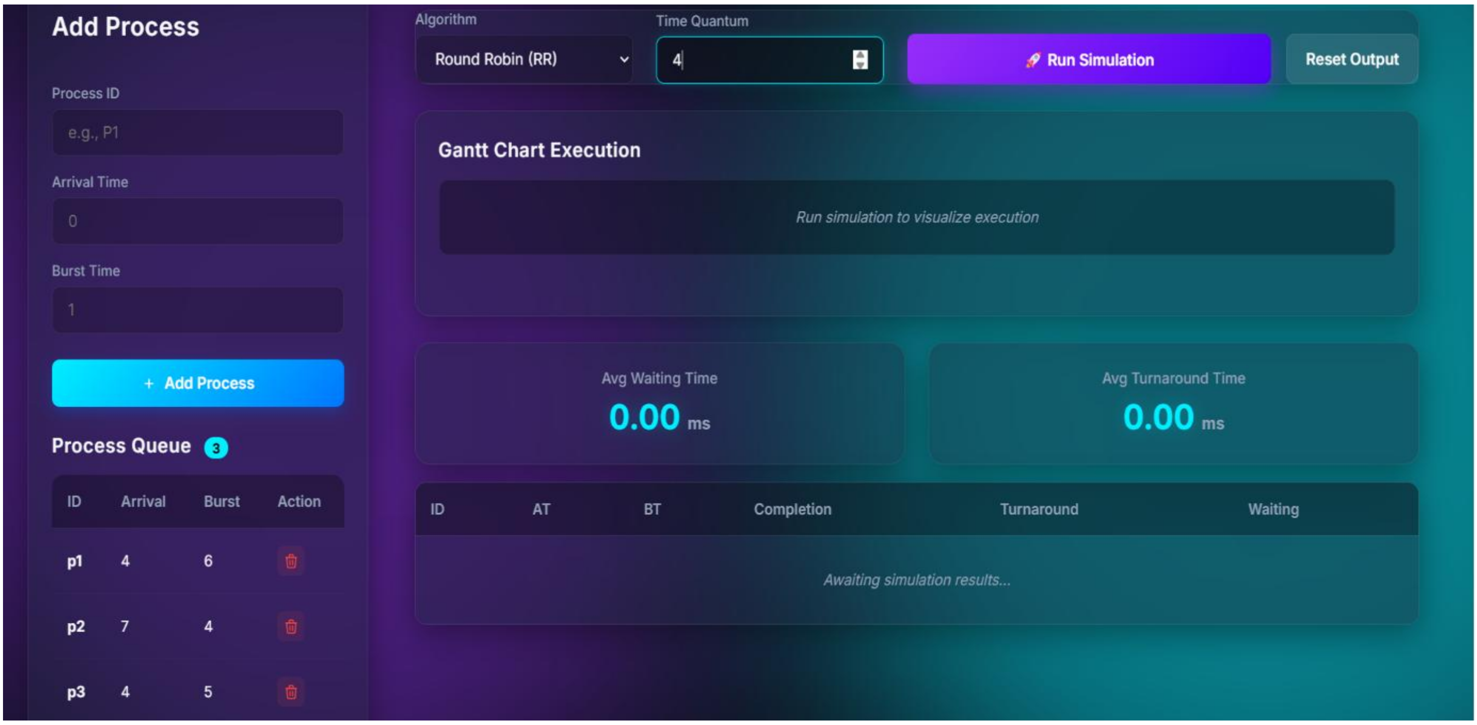
Step 4: Adding Third Process (P3)

The third process **P3** is added with Arrival Time (4) and Burst Time (5). At this stage, the process queue contains all required inputs for performing scheduling.



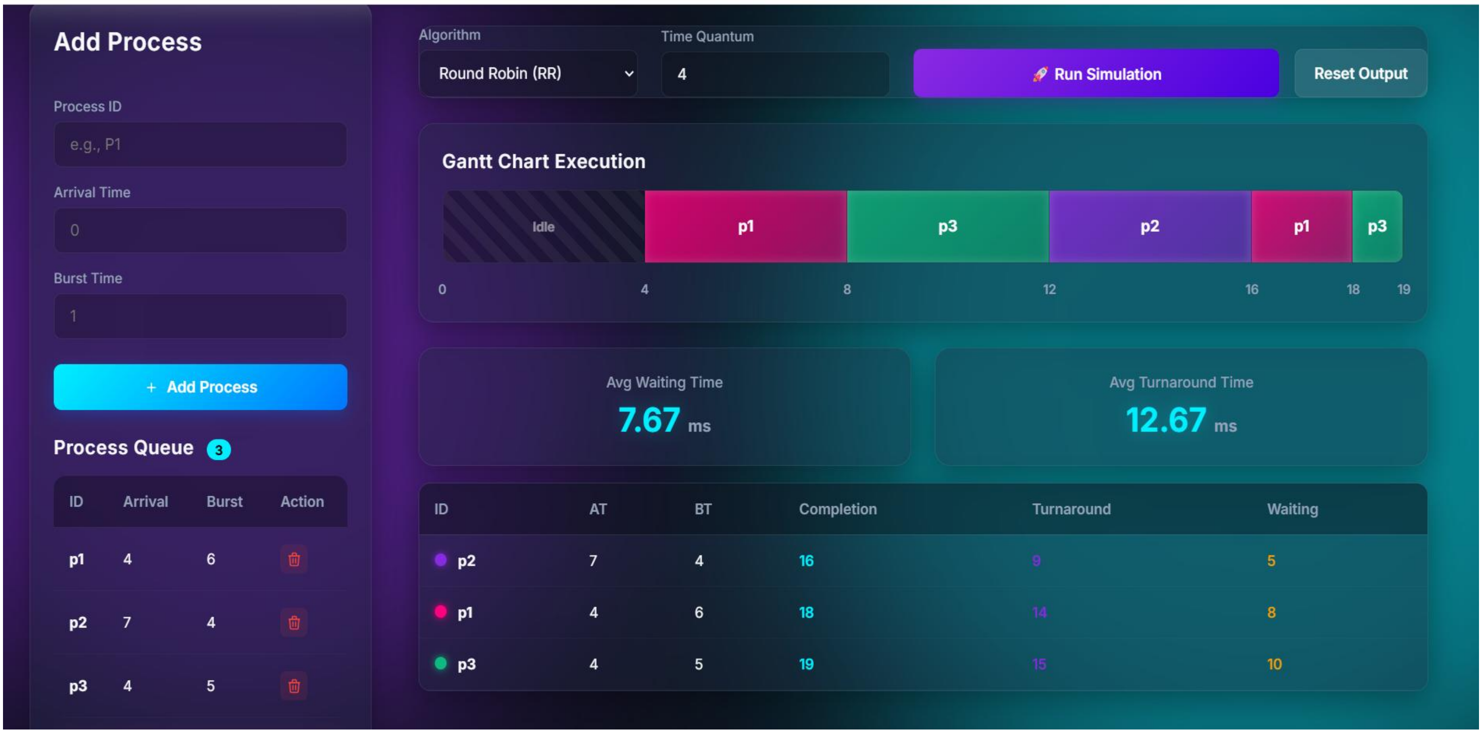
Step 5: Selecting Algorithm and Running Simulation

Here, the user selects the **Round Robin (RR)** scheduling algorithm and sets the Time Quantum to 4. The simulation is then executed by clicking the "Run Simulation" button.



Step 6: Gantt Chart and Output Visualization

This step displays the final output of the simulation. The Gantt Chart visually represents the execution order of processes over time. Additionally, performance metrics such as Average Waiting Time and Average Turnaround Time are calculated and displayed. A detailed table shows completion time, turnaround time, and waiting time for each process.



The CPU Scheduling Visualizer effectively demonstrates the working of scheduling algorithms in a clear and interactive way. It helps users understand process execution, waiting time, and turnaround time through visual representation. One major advantage of this system is its ability to simplify complex scheduling concepts using Gantt charts. However, the accuracy of results depends on correct input values and limited algorithm support can be considered a minor drawback.